

<i>Title:</i>	Revision of the Bachelor of Science in Chemical Engineering
<i>Sponsor:</i>	<i>Department of Chemical Engineering</i>
<i>Executive Summary:</i>	There are currently 18 required courses in the College of Engineering that comprise the BS in Chemical Engineering. One of these courses is CHE 396: Senior Design I. The department proposes to decrease the credit hours of this course from 4 to 3 hours to reflect the course load/level of effort required by students. In addition, changes to coursework in chemistry are reducing the hours of nonengineering coursework in the degree by two hours. Further, the department proposes to add a 3-credit hour course to the BS in Chemical Engineering degree requirement: CHE 230 Molecular Systems in Chemical Engineering. CHE 230 was approved as a new course in Spring 2020. As a result, the total hours required to earn the degree will remain at 128.
<i>Description:</i>	In the Chemical Engineering curriculum, students learn to apply chemistry, physics, and mathematics to the industrial-scale design and production of materials and chemicals, including petroleum products, polymers, pharmaceuticals, electronic devices, consumer products, and foods. This program also explores chemical engineering applications in developing processes for environmental protection, waste treatment and reducing global warming gases, the creation of alternative energy sources, and other frontiers, such as biotechnology, drug delivery, molecular engineering, microelectronic materials and nanotechnology. The BS in Chemical Engineering, like all engineering degrees in the college, requires 128 hours in total. This is comprised of 70 hours of nonengineering and General Education requirements, 49 hours of coursework in the College of Engineering, 3 hours of a polymer science or physical chemistry II selective, 3 hours of a technical elective in chemical engineering, and 3 hours of electives outside of the major rubric. The Department of Chemical Engineering proposes to revise the BS degree as follows: 1. Decrease the credit hours for CHE 396: Senior Design I from 4 to 3 hours. 2. The credit hours of CHEM 232: Organic Chemistry I and CHEM 234 Organic Chemistry II are being reduced by one hour each (from 4 to 3 hours). These changes were part of a recent proposal approved by the Senate on January 27, 2022. As a result, the hours of nonengineering and General Education coursework in the degree will decrease by two hours (from 70 to 68 hours). 3. Add CHE 230: Molecular Systems in Chemical Engineering (3 hours) as a degree requirement. Given the reduction in hours to CHE 396, the addition of CHE 230 will result in a change to hours required in the college from 49 to 51 hours.

4. Reflect the other changes to courses offered by the Department of Chemistry.
 - a. CHEM 122: General Chemistry I Lecture (4 hours), and CHEM 123: General Chemistry Laboratory I (1 hour) will have their prerequisites updates so that the two courses no longer must be taken concurrently. CHEM 122 will be retitled "Matter and Energy" and offered as a 3-hour course. CHEM 123 will be retitled "Foundations of Chemical Inquiry I" and offered as a 2-hour course.
 - b. CHEM 124: General Chemistry II Lecture (4 hours), and CHEM 125: General Chemistry Laboratory II (1 hour) will have their prerequisites updates so that the two courses no longer must be taken concurrently. CHEM 124 will be retitled "Chemical Dynamics" and offered as a 3-hour course. CHEM 125 will be retitled "Foundations of Chemical Inquiry II" and offered as a 2-hour course.
 - c. As noted above, CHEM 232: Organic Chemistry (4 hours) will become a 3-hour course. It will also be retitled "Structure and Function."
 - d. CHEM 233: Organic Chemistry Laboratory I (2 hours) will be retitled "Synthetic Techniques Laboratory."
 - e. As noted above, CHEM 234: Organic Chemistry II (4 hours) will become a 3-hour course. It will also be retitled "Chemical Synthesis."

As a result of these changes, the total hours will remain 128. But the distribution of hours will change as noted in the table below.

	Current Hours	Proposed Hours
Nonengineering/GenEd	70	68
Required in College	49	51
Selective	3	Same
Technical Elective	3	Same
Elective Outside Major	3	Same
Total Hours	128	Same

Justification:

1. CHE 396: Senior Design I is the first course of the two-semester capstone chemical engineering senior design course sequence (CHE 396/397); recent improvements in the course have decreased the course load and justifies a 3-credit hour course. The changes organize the topics that subsequently are used in 397 in a more orderly fashion to provide introductions and allow in-class projects and presentations that are completed within the lecture period.
2. CHE 230: Molecular Systems in Chemical Engineering is designed to provide an introduction to molecular engineering and material science in relation to chemical engineering. It covers both materials chemistry and physics of bulk materials and nanomaterials for a broad range of applications and is an important addition in keeping the chemical engineering curriculum up-to-date.
3. Due to the change in CHE 396 credit hours, addition of CHE 230, and change in CHEM 232 and CHEM 234 credit hours, the Total Hours Required in COE will increase and the Non-Eng/GenEd requirements will decrease.
4. Due to approved changes in CHEM courses from LAS, the undergraduate catalog needs to be revised to reflect these changes.

<i>Catalog Statement:</i>	<i>See below</i>
<i>Minority Impact Statement:</i>	5. The changes will clarify the degree requirements for all students and will have a positive impact on minority students. Changes in 396 organize the topics in an orderly fashion so that the students can easily utilize them in 397. CHE 230 is at the forefront of a new direction in chemical engineering; molecular engineering and will keep the students up-to-date on new topics.
<i>Budgetary and Staff Implications:</i>	None anticipated. Faculty size has increased with a newly appointed professor who is currently teaching the course. Also, one section is sufficient for accommodating all the BS in Chemical Engineering majors; at most there will be 60-65 students taking the course each year.
<i>Library Resource Implications:</i>	<i>None anticipated as all courses are currently part of the department curriculum.</i>
<i>Space Implications:</i>	<i>Additional sections are not anticipated for CHE 230</i>
<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	Provide unit dates, including department, college/school committees, faculty Approvals, as applicable. Chemical engineering faculty approval: 08/03/2020 Chemical Engineering department approval: 08/03/2020 Education Policy Committee approval: 11/18/2020
<i>Contact Person:</i>	Alan Zdunek zdunek@uic.edu
<i>Proposed Effective Date/Term:</i>	Fall 2022

Current	Revised
BS in Chemical Engineering	BS in Chemical Engineering
<i>Degree Requirements</i> To earn a Bachelor of Science in Chemical Engineering degree	SAME

from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students should consult the *College of Engineering* section for additional degree requirements and college academic policies.

BS in Chemical Engineering Degree Requirements	Hours
Nonengineering and General Education Requirements	70
Required in the College of Engineering	49
Selective ^a	3
Technical Elective ^a	3
Electives outside the Major Rubric ^a	3
Total Hours—BS in Chemical Engineering	128

^a Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.

Nonengineering and General Education Requirements

ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3
ENGL 161— Academic Writing II: Writing for Inquiry and Research	3
Exploring World Cultures course ^a	3
Understanding the Creative Arts course ^a	3
Understanding the Past course ^a	3
Understanding the Individual and Society course ^a	3
Understanding U.S. Society course ^a	3
MATH 180—Calculus I ^b	4
MATH 181—Calculus II ^b	4
MATH 210—Calculus III ^b	3
MATH 220—Introduction to Differential Equations	3
PHYS 141—General Physics I (Mechanics) ^b	4
PHYS 142—General Physics II (Electricity and Magnetism) ^b	4
CHEM 122- General College Chemistry I ^b (Lecture)	4
CHEM 123- General College Chemistry I ^b (Laboratory)	1
<u>OR</u> CHEM 116 – Honors and Major General and Analytical Chemistry I	5
CHEM 124—General College Chemistry II ^b (Lecture)	4
CHEM 125—General College Chemistry II ^b (Lab)	1
<u>OR</u> CHEM 118 – Honors and Major General and Analytical Chemistry II	5
CHEM 222—Analytical Chemistry ^c	4
CHEM 232—Organic Chemistry I	4
CHEM 233- Organic Chemistry Laboratory	2
CHEM 234—Organic Chemistry II	4
CHEM 342—Physical Chemistry I	3
Total Hours- Nonengineering and General Education Requirements	70

^a Students should consult the General Education section of the catalog for a list of approved courses in this category.

^b This course is approved for the Analyzing the Natural World General Education category.

^c Students who take CHEM 116 and CHEM 118 to fulfill the general chemistry requirement do not need to take CHEM 222.

BS in Chemical Engineering Degree Requirements	Hours
Nonengineering and General Education Requirements	68
Required in the College of Engineering	51
Selective	3
Technical Elective ^a	3
Electives outside the Major Rubric ^a	3
Total Hours—BS in Chemical Engineering	128

^a Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.

Nonengineering and General Education Requirements

ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3
ENGL 161— Academic Writing II: Writing for Inquiry and Research	3
Exploring World Cultures course ^a	3
Understanding the Creative Arts course ^a	3
Understanding the Past course ^a	3
Understanding the Individual and Society course ^a	3
Understanding U.S. Society course ^a	3
MATH 180—Calculus I ^b	4
MATH 181—Calculus II ^b	4
MATH 210—Calculus III ^b	3
MATH 220—Introduction to Differential Equations	3
PHYS 141—General Physics I (Mechanics) ^b	4
PHYS 142—General Physics II (Electricity and Magnetism) ^b	4
CHEM 122- Matter and Energy^b	3
CHEM 123- Foundations of Chemical Inquiry I^b	2
<u>OR</u> CHEM 116 – Honors and Major General and Analytical Chemistry I	5
CHEM 124— Chemical Dynamics^b	3
CHEM 125— Foundations of Chemical Inquiry II^b	2
<u>OR</u> CHEM 118 – Honors and Major General and Analytical Chemistry II	5
CHEM 222—Analytical Chemistry ^c	4
CHEM 232—Structure and Function	3
CHEM 233- Synthetic Techniques Laboratory	2
CHEM 234—Chemical Synthesis	3
CHEM 342—Physical Chemistry I	3
Total Hours- Nonengineering and General Education Requirements	68

^a Students should consult the General Education section of the catalog for a list of approved courses in this category.

^b This course is approved for the Analyzing the Natural World General Education category.

Instead they should enroll in one of the following: CHEM 314: Inorganic Chemistry; CHEM 452: Biochemistry; or CHEM 402: Chemical Information Systems and CHEM 444: Physical Chemistry III.		^c Students who take CHEM 116 and CHEM 118 to fulfill the general chemistry requirement do not need to take CHEM 222. Instead they should enroll in one of the following: CHEM 314: Inorganic Chemistry; CHEM 452: Biochemistry; or CHEM 402: Chemical Information Systems and CHEM 444: Physical Chemistry III.	
Required in the College of Engineering		Required in the College of Engineering	
Courses	Hours	SAME	
ENGR 100—Orientation ^a	0 ^a		
CHE 201—Introduction to Thermodynamics	3		
CHE 205—Computational Methods in Chem. Eng.	3		
CHE 210—Material and Energy Balances	4		
CHE 301—Chemical Engineering: Thermodynamics	3		
CHE 311—Transport Phenomena I	3		
CHE 312—Transport Phenomena II	3	Add, CHE 230—Mol Sys Chem Eng	
CHE 313—Transport Phenomena III	3	3	
CHE 321—Chemical Reaction Engineering	3		
CHE 341—Chemical Process Control	3		
CHE 381—Chemical Engineering Laboratory I	2		
CHE 382—Chemical Engineering Laboratory II	2		
CHE 396—Senior Design I	4		
CHE 397—Senior Design II	4		
CHE 499—Professional Development Seminar	0		
ECE 210—Electrical Circuit Analysis	3	Reduce hours of, CHE 396—Senior Design II	
CME 260—Properties of Materials	3	3	
CS 109—Programming for Engineers with MatLab	3		
Total Hours—Required in the College of Engineering	49	Total Hours—Required in the College of Engineering	51
a ENGR 100 is one-semester-hour course, but the hour does not count toward the total hours required for graduation.		SAME	
Selective			
Courses		SAME	
Select one of the following:	3		
CHE 330 Polymer Science			
CHEM 346 Physical Chemistry II			
Total Hours	3		
Technical Elective			
Courses	Hours		
One technical elective to be chosen from the following list of design-oriented courses:^a			
CHE 392—Undergraduate Research (3) ^b	3		

CHE 413—Introduction to Flow in Porous Media (3) CHE 421—Combustion Engineering (3) CHE 422—Biochemical Engineering (3) CHE 423—Catalytic Reaction Engineering (3) CHE 425 Nanotechnology Drug Deliver (3) CHE 433 Process Simulation with Aspen Plus (3) CHE 438—Computational Molecular Modeling (3) CHE 440—Non-Newtonian Fluids (3) CHE 441—Computer Applications in Chemical Engineering (3) CHE 450—Air Pollution Engineering (4) CHE 451 Renewable Energy Technologies (3) CHE 453 Fundamentals of Electrochemistry (3) CHE 454 Molecular and Macromolecular Engineering (3) CHE 455 Nanoscale Systems in ChE (3) CHE 456—Fundamentals and Design of Microelectronics Processes (3) CHE 457 Colloidal and Interfacial Phenomena (3) CHE 494—Selected Topics in Chemical Engineering (3) Total Hours—Technical Elective 3 a Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee. b An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering. Electives outside the Major Rubric <table border="1"> <thead> <tr> <th colspan="2">Electives</th></tr> </thead> <tbody> <tr> <td>Electives outside the CHE rubric</td><td>3</td></tr> <tr> <td>Total Hours</td><td>3</td></tr> </tbody> </table> Optional Concentrations BS in Chemical Engineering, Biochemical Engineering Concentration BS in Chemical Engineering, Energy and Environment Concentration BS in Chemical Engineering, Entrepreneurship Concentration BS in Chemical Engineering, Nanotechnology Concentration BS in Chemical Engineering, Polymers and Molecular Engineering Concentration BS in Chemical Engineering, Process Automation Concentration	Electives		Electives outside the CHE rubric	3	Total Hours	3	SAME
Electives							
Electives outside the CHE rubric	3						
Total Hours	3						

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Sample Course Schedule

Freshman Year		
FIRST SEMESTER		
ENGR 100	Engineering Orientation ^a	
ENGL 160	Academic Writing I: Writing in Academic and Public Contexts	
MATH 180	Calculus I	4
CHEM 122 & CHEM 123 or CHEM 116	Matter and Energy & Foundations of Chemical Inquiry I or Honors and Majors General and Analytical Chemistry I	5
General Education Core Course		3
Hours		15
SECOND SEMESTER		
CS 109	Programming for Engineers with MatLab	3
MATH 181	Calculus II	4
PHYS 141	General Physics I (Mechanics)	4
CHEM 124 & CHEM 125 or CHEM 118	Chemical Dynamics & Foundations of Chemical Inquiry II or Honors and Majors General and Analytical Chemistry II	5
Hours		16

Commented [WDC1]: Alan, it was too difficult to insert the sample schedule in the side-by-side table above. So I added it to the bottom of the proposal, and edits are not made using tracked changes (but the strikethrough feature, text in bold, etc.

Please review and let me know if there are edits that need to be made. Where should CHE 230 be added?

Commented [ZA2R1]: Made the changes. I also added CHE 230 in Sophomore Year Second Semester

Sophomore Year		
FIRST SEMESTER		
ENGL 161	Academic Writing II: Writing for Inquiry and Research	3
PHYS 142	General Physics II (Electricity and Magnetism)	4
CHE 201	Introduction To Thermodynamics	3
MATH 210	Calculus III	3
CHEM 222	Analytical Chemistry	4
Hours		17
SECOND SEMESTER		
CHE 205	Computational Methods in Chemical Engineering	3
CHE 210	Material and Energy Balances	4
CHE 230	Molecular Systems in Chemical Engineering	3
MATH 220	Introduction to Differential Equations	3
CHEM 232	Structure and Function	4 3
CHEM 233	Synthetic Techniques Laboratory	2
Hours		46 18
Junior Year		
FIRST SEMESTER		
CHE 301	Chemical Engineering Thermodynamics	3
CHE 311	Transport Phenomena I	3
CHEM 234	Chemical Synthesis	4 3
CHEM 342	Physical Chemistry I	3
ECE 210	Electrical Circuit Analysis	3
Hours		46 15
SECOND SEMESTER		
CHE 312	Transport Phenomena II	3
CHE 313	Transport Phenomena III	3
CHE 321	Chemical Reaction Engineering	3
CME 260	Properties of Materials	3
General Education Core Course		3
Hours		15
Senior Year		

FIRST SEMESTER		
CHE 381	Chemical Engineering Laboratory I	2
CHE 396	Senior Design I	4 3
CHE 499	Professional Development Seminar	0
General Education Core course		3
General Education Core Course		3
CHE Technical Elective (select from list)		3
CHE 330	Polymer Science (Or CHEM 346 Junior Year second semester)	3
Hours		18 17
SECOND SEMESTER		
CHE 341	Chemical Process Control	3
CHE 382	Chemical Engineering Laboratory II	2
CHE 397	Senior Design II	4
Elective outside the Major Rubric		3
General Education Core course		3
Hours		15
Total Hours		128
Plan of Study Grid		

^a [ENGR 100](#) is a one-semester-hour course, but the hour does not count toward the total hours required for graduation.